SoulGrad: Designing an LLM Companion for Mental Wellbeing in Postgraduate Exam Preparation in China

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Abstract

Candidates preparing for China's national postgraduate entrance examination typically spend six to twelve months in solitary study with limited structured support, facing anxiety, isolation, and unstable motivation that undermine sustained engagement. Existing tools either track productivity without addressing emotional needs, or offer reactive support that fails to sustain long-term engagement. To address this gap, we conducted a mixed-method formative study ($N = 93$ surveys, $N = 15$ interviews) and identified four structural challenges: a black-box preparation ecology, anxiety amplified by zero-sum peer comparison, motivational collapse as daily stressors compound, and distrust toward existing LLM-based systems due to shallow personalization. Guided by these findings, we built SoulGrad, a computational companion that integrates behavioral sensing with psychologically informed support through proactive attributional intervention, non-competitive co-presence, expressive modalities, and user-controlled sensing. A 16-day longitudinal pilot study with 16 participants showed positive trends in emotional stability and anxiety reduction, with encouraging signs of strategy transfer beyond the study context. We reflect on the ethical tensions of deploying LLM-driven support for vulnerable populations and contribute recommendations for responsible integration in mental health systems.

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CCS Concepts

• **Human-centered computing → Empirical studies in HCI**.

Keywords

High-Pressure Learning, Exam Preparation, Mental Health Support, Proactive Intervention, Behavior-Aware Systems

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1 Introduction

China's postgraduate entrance examination attracts over 3.4 million candidates annually [26]. Candidates typically spend six to twelve months in solitary, self-directed study, cut off from structured feedback and peer networks, resulting in chronic anxiety, deep isolation, and pervasive uncertainty [18], often manifesting as sleep disorders, motivational collapse, or clinical depression. This context demands everyday computing systems that can sense, interpret, and respond to mental health needs over months [3].

Existing tools address only fragments of this challenge. Productivity trackers monitor study behavior but ignore emotional state. Conversational agents can reduce short-term anxiety [1, 8], yet LLM-based systems often suffer from shallow personalization, interaction fatigue, and trust erosion [13]. Just-in-time adaptive interventions target immediate stressors rather than the cumulative breakdowns that characterize months-long preparation [15]. In high-stakes exam preparation, cognitive performance and emotional wellbeing form a tightly coupled dynamic: a single setback can spiral into rumination, exhaustion breeds guilt about resting,

and emotional fluctuations directly impair learning. No existing system actively mediates this cycle over time.

To address this gap, we investigate two research questions: **RQ1.** *What mental health challenges do candidates experience during postgraduate exam preparation, and what expectations do they hold for technological support?* **RQ2.** *How can a computational companion system that couples behavioral sensing with psychological intervention be designed, and how does it influence wellbeing and engagement over the study period?*

To answer these questions, we first conducted a mixed-method formative study ($N = 93$ surveys, $N = 15$ interviews) to understand the lived experiences of postgraduate exam candidates. Our analysis identifies four structural challenges: black-box isolation, zero-sum peer comparison, cascading motivational breakdowns, and distrust toward existing systems due to shallow personalization. These findings reveal that candidates' struggles are structurally embedded rather than episodic. Guided by these insights, we derived four design guidelines and built SoulGrad, a computational companion that tightly integrates behavioral sensing with proactive attributional intervention, non-competitive co-presence, expressive modalities, and user-controlled sensing. We evaluated SoulGrad in a 16-day longitudinal pilot study with 16 participants, and the results showed positive trends across multiple psychological dimensions. In sum, this work makes three contributions:

(1) A mixed-method formative study (93 surveys, 15 interviews) identifying four structural challenges that reveal candidates' struggles are structurally embedded rather than episodic.
(2) Four design guidelines and their instantiation in SoulGrad, a computational companion coupling behavioral sensing with psychologically informed intervention.
(3) A 16-day longitudinal study with 16 participants showing positive trends in emotional stability, anxiety, and resilience, with evidence of strategy transfer.

2 Related Work

We review prior work in two broad areas: agent-based support and JITAIs for mental health, and context-aware sensing with privacy and trust.

2.1 Agent-Based Support and JITAIs

Conversational agents have shown promise in reducing depression and anxiety, with Woebot demonstrating efficacy in randomized trials [8]. Meta-analyses confirm positive effects across both rule-based and generative chatbots [1], and culturally adapted versions have recently been developed for university populations. However, these systems remain fundamentally reactive: they offer either emotional comfort or cognitive tutoring, but rarely integrate both within a single interaction flow. Moreover, emerging work on LLM-based mental health prediction [25] raises concerns about hallucination risk when these models are deployed in clinical contexts [9].

JITAIs deliver support at vulnerable moments through continuous sensing and decision rules [15]. Yet JITAIs face persistent disengagement in high-anxiety populations [14, 17] and target immediate stressors rather than cumulative breakdowns.

2.2 Context-Aware Sensing, Privacy, and Trust

Longitudinal mobile sensing can capture mental health trajectories. MindScape integrates behavioral sensing with LLMs for journaling but remains reflective rather than proactive [16]. In high-pressure populations, users reject facial recognition but accept heart rate monitoring during designated windows [11]. Disengagement is driven by perceived algorithmic predictability and lack of transparent boundaries [17]. No existing system simultaneously couples longitudinal sensing with proactive psychological support, provides non-competitive co-presence, and respects absolute privacy red lines. SoulGrad addresses this gap while serving as a case study for examining LLM integration risks in mental health contexts.

3 Formative Study

To understand candidates' lived experiences, we conducted a mixed-method formative study combining surveys and interviews.

3.1 Methods

We employed a mixed-methods approach approved by our university IRB. All participants provided informed consent.

Online Questionnaires. We collected demographics, psychological states (GAD-7 [22]), exam-specific burdens (five custom items on social comparison, catastrophizing, sunk-cost fear, and isolation), satisfaction with existing tools, and expectations for LLM-driven technological support. We received 93 valid responses spanning diverse preparation stages (83.9% preparing, 12.9% admitted, 3.3% failed/abandoned) and disciplines (Fig. 1).

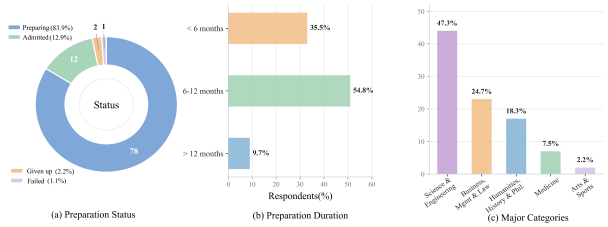


Figure 1: Demographic profile of 93 survey respondents across (a) preparation status, (b) duration, and (c) major categories.

Semi-structured Interviews. We purposively selected 15 participants (11 female, 4 male; $M_{age} = 22.2$, $SD = 1.26$) from the survey pool, spanning STEM, Business, Humanities, and Medicine. Interviews (40–60 min) covered four themes: motivation and stage-specific needs, embodied distress, limitations of existing support, and participatory system design. We used reflexive thematic analysis [4] with inter-coder reliability $\kappa = 0.87$.

3.2 Findings and Design Guidelines

We synthesized four themes characterizing candidates' structural challenges, then derived four design guidelines (DGs) mapped to these findings (Table 1).

F1: Black-Box Isolation and Cascading Breakdown. 82.8% of participants frequently experienced isolation. Lacking external scaffolding, candidates resort to defensive socializing driven by

Table 1: Mapping Empirical Findings to Design Guidelines.

Empirical Findings	Design Guidelines (DGs)
F1: Motivational vulnerability, relaxation guilt, cascading breakdown. F2: LLM pseudo-empathy. F3: Need for attributional restructuring.	**DG1**: Proactive, context-anchored intervention replaces passive comfort.
F1: Structural loneliness vs. zero-sum peer pressure. F2: Reciprocity burden in human comfort.	**DG2**: De-toxified co-presence and low-involvement emotional receptacles.
F4: Self-censorship with text; acceptance of unobtrusive sensing.	**DG3**: Implicit sensing and expressive modalities bypass psychological defenses.
F3: Fear of LLM hallucinations; need for traceability. F4: Absolute red lines on visual surveillance.	**DG4**: Epistemic trust through transparent boundaries and user agency.

zero-sum competition. Micro-stressors cascade into systemic breakdowns: a mother opening the door late triggers emotional sensitization (P5), leading to insomnia, anorexia, and binge eating (P9, P14). Relaxation guilt prevents recovery (P7).

F2: The Intervention Gap of Existing LLM-based Tools. 87.1% had used LLM-based chatbots, but satisfaction was moderate ($M = 4.03/7$). Candidates rejected what they described as the *pseudo-empathy* of generic LLMs: algorithmically predictable responses that felt patronizing (P13, P15) and personas that seemed overly people-pleasing (P14). Participants also noted that the lack of long-term memory in existing chatbots caused a sense of relationship rupture, where the assistant failed to remember past conversations and thus could not build sustained rapport (P13).

F3: Demand for Integrated Academic-Psychological Support. Candidates demanded deep binding of academic diagnostics with emotional intervention (P12). They preferred proactive attributional interventions (P5) and insisted on traceable advice to combat hallucination fear (P4).

F4: Privacy Calculus and Modality Preferences. Participants established an absolute defense line against facial biometrics (P14) and visual surveillance (P1), but accepted smartwatch heart rate instead [11]. They demanded absolute autonomy over intervention timing (P1) and preferred voice interaction for lowering emotional disclosure barriers.

4 System Design and Implementation

SoulGrad operationalizes the four DGs in an integrated architecture (Fig. 2).

4.1 DG1: Proactive Attributional Intervention

SoulGrad computes a composite daily stress score S_t via normalized weighted fusion of six indicators:

$$S_t = \sum_{i=1}^{6} w_i \cdot \frac{x_{i,t} - \mu_i}{\sigma_i} \tag{1}$$

where $x_{i,t}$ is the raw value of indicator i on day t, and (μ_i, σ_i) are the population mean and standard deviation. The six indicators and weights are: heart rate variability ($w_1 = 0.25$), daily learning duration ($w_2 = 0.20$), task interruption frequency ($w_3 = 0.15$), task completion rate ($w_4 = 0.15$), error proportion ($w_5 = 0.15$), and sleep duration ($w_6 = 0.10$). Each is standardized using a 7-day rolling z-score to account for individual baselines. The composite score determines four hierarchical warning levels (Table 2), with thresholds heuristically set based on formative study patterns (F1).

At L3–L4, the system triggers a three-step intervention grounded in Weiner's Attribution Theory [23]: (1) *Emotional validation*, acknowledging frustration; (2) *Cognitive reappraisal*, reframing setbacks from internal-stable to external-unstable attributions; and (3) *Micro-step suggestions* to rebuild positive feedback loops. A *front-loaded buffer* sets the target completion date 2–3 weeks before the actual exam [12], and phase-aware regulation shifts emphasis from academic planning to affective support as the exam approaches (Fig. 3).

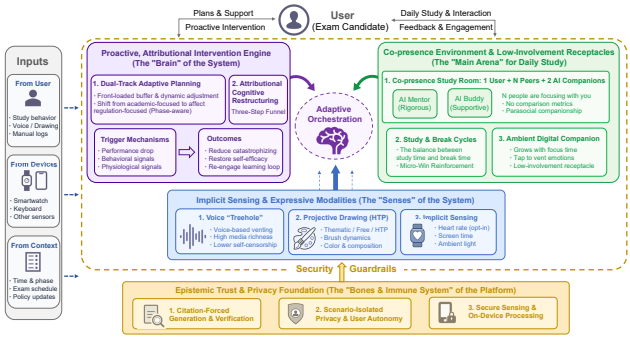


Figure 2: SoulGrad system architecture. DG1: central orchestration engine; DG2: co-presence learning environment; DG3: multimodal sensing; DG4: privacy and trust foundations.

Table 2: Hierarchical Intervention Thresholds.

Level	S_t Range	System Response
L1 (Normal)	< 0.5	Passive monitoring only
L2 (Elevated)	$[0.5, 1.0)$	Gentle check-in
L3 (High)	$[1.0, 2.0)$	Proactive attributional intervention
L4 (Critical)	≥ 2.0	Prescribed break + safety check

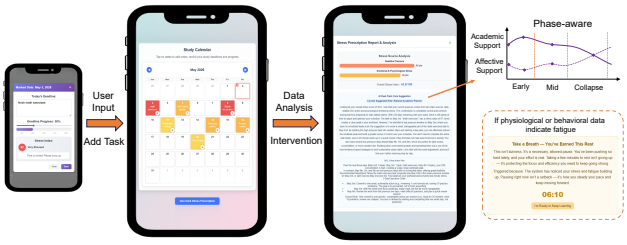


Figure 3: From deadline input to stress-aware replanning: task capture, workload visualization, and system-driven redistribution with buffer insertion.

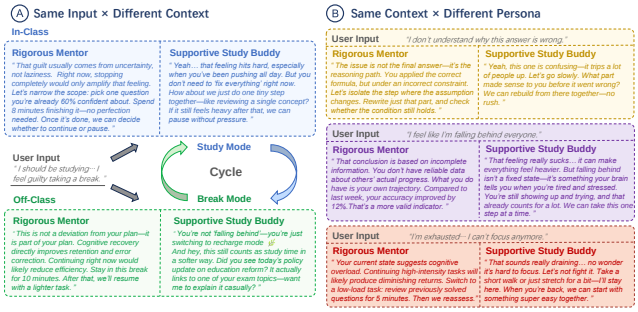


Figure 4: Context- and persona-aware interaction patterns across study and break modes.

4.2 DG2: De-toxified Co-presence

SoulGrad creates a **1+N+2** space: one user, N anonymized peers, and two system companions. Only crowd-level prompts are shown (e.g., *N people are focusing with you*); all personal progress remains hidden. Drawing on parasocial interaction theory [10], this provides collective companionship without zero-sum comparison. The two built-in companions can switch between different interaction styles (e.g., a rigorous mentor or an empathetic study partner), adapting their tone to the user's current context. When continuous focus exceeds 90 min, the system triggers a prescribed 15-minute break with casual conversation blending exam-related topics with personal interests (Fig. 4).

An **Ambient Digital Companion**, visualized as a digital plant, grows gradually with the user's cumulative focus time, providing a calming visual presence for attention restoration. Users can tap the plant at any time to vent raw emotions; the companion absorbs these expressions without generating a textual reply, producing only subtle visual or auditory feedback. This design provides a low-involvement emotional receptacle that mitigates the reciprocity burden associated with human-to-human comfort (F2).

4.3 DG3: Implicit Sensing and Expressive Modalities

Voice Treehole. SoulGrad provides voice-based venting that maintains a warm, non-directive presence without semantic analysis. High-fidelity voice delivery strengthens perceived co-presence and bypasses rational self-censorship (F4).

Projective Drawing Assessment. A digital House-Tree-Person (HTP) module captures brush dynamics, color, composition, and symbolic elements. Computer vision and LLM-based analysis generate gamified psychological portraits rather than clinical diagnoses, transforming anxiety into concrete artifacts that encourage repeated engagement.

4.4 DG4: Epistemic Trust and Safety

Citation-Forced Generation. All academic guidance carries inline citations linked to verifiable sources. A lightweight verification agent checks claims via retrieval-augmented generation; if confidence is below 0.9, the system declines to answer.

Scenario-Isolated Privacy. Heart-rate monitoring is enabled only during user-specified study hours. Facial data is never collected. A prominent *Do Not Disturb* button immediately stops all proactive interventions. All raw data remain on-device by default; cloud processing uses differential privacy.

Safety Guardrails. Given the vulnerability of our target population, we implemented multiple safety layers: (1) a content classifier screens generated text; (2) a crisis protocol redirects to professional helplines when risk signals are detected; (3) LLM-generated psychological content is constrained to evidence-based frameworks; and (4) users can disable the LLM companion while retaining study planning features.

5 User Study

We evaluated SoulGrad through a 16-day longitudinal field deployment to assess its real-world performance and user experience within a high-stakes exam preparation context.

5.1 Study Design

We conducted a 16-day longitudinal pilot study with a four-phase within-subjects design (Baseline → Phase 1: Basic scheduling → Phase 2: Cognitive intervention → Phase 3: Full system). Due to the absence of a control group and the small sample size, this study serves as a feasibility pilot evaluation rather than a confirmatory trial.

Participants. Sixteen full-time students (M_{age} = 22.4, SD = 1.5, range 19–25) completed the 16-day protocol. Majors: Science & Engineering (n = 8), Economics/Management/Law (n = 5), Humanities (n = 3). Preparation duration: <6 months (n = 5), 6–12 months (n = 9), >12 months (n = 2). Baseline GAD-7: M = 10.2, SD = 3.8 (mild to moderate anxiety). Participants were blind to phase boundaries. All procedures received IRB approval.

Measures. We administered four validated psychometric scales: GAD-7 [22] and STAI [21] for anxiety; GSES [20] for self-efficacy; MBI-SS [19] for burnout; and CD-RISC-10 [6] for resilience (all using official Chinese versions). Measures were collected at the end of each phase to reduce reactivity; full STAI was administered only at baseline and post-intervention. System evaluation employed SUS [5], TAM [7], and UES-SF [24] after each phase. Exit interviews (30–45 min) were analyzed via reflexive thematic analysis [4] (κ = 0.87).

5.2 Results

Psychometric Outcomes. Repeated-measures ANOVA revealed trends toward improvement across all measures (Table 3), though the staged design without a control group means these changes cannot be causally attributed to the intervention alone. GAD-7 decreased from baseline (M = 10.19) to Phase 3 (M = 5.31; η_p^2 = 0.49). STAI state anxiety showed a similar pattern (d = 1.21). MBI-SS burnout decreased (η_p^2 = 0.53), with reductions in emotional exhaustion (32.4%) and cynicism (38.7%). Self-efficacy (η_p^2 = 0.31) and resilience (η_p^2 = 0.43) also improved (baseline-to-Phase 3 d = 0.65 and d = 0.96). We did not correct for multiple comparisons, and all p-values are exploratory; effect sizes are reported to inform future power analyses rather than to establish efficacy.

Box plots of psychometric measures are presented in Fig. 5.

System Evaluation. SUS scores remained below the standard benchmark of 68 throughout the study (Table 4), dipping in Phase

Table 3: Mean (SD) Psychometric Measures by Phase ($N = 16$).

Scale	Baseline	Phase 1	Phase 2	Phase 3	F	η_p^2
GAD-7 (0–21)	10.19 (3.82)	8.44 (4.02)	6.81 (3.57)	5.31 (2.96)	14.27***	0.49
GSES (10–40)	28.38 (5.21)	29.56 (5.44)	31.31 (5.13)	32.44 (4.62)	6.89**	0.31
CD-RISC-10 (0–40)	30.63 (5.07)	31.88 (5.11)	34.06 (4.68)	35.31 (4.35)	11.23***	0.43
MBI-SS Total	51.44 (9.23)	46.81 (8.46)	42.13 (7.19)	38.94 (6.67)	16.83***	0.53

*** $p < .001$, ** $p < .01$ (uncorrected).

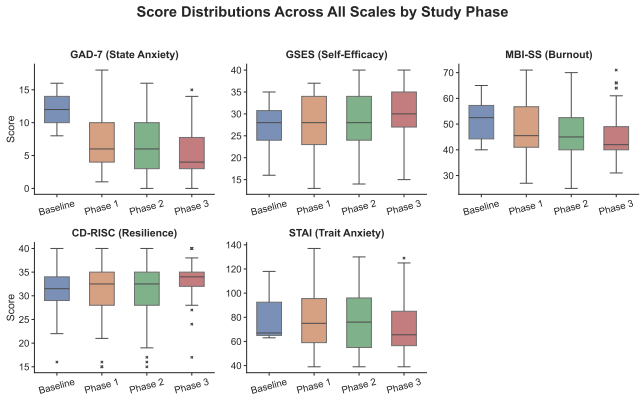


Figure 5: Score distributions for all psychometric measures by study phase.

2 ($M = 49.38$) when cognitive interventions were introduced, then partially recovering in Phase 3 ($M = 54.69$). This pattern suggests that adding proactive intervention features introduced non-trivial usability friction, consistent with the qualitative finding that some participants initially found attributional restructuring challenging. However, TAM usefulness and UES-SF reward increased steadily across phases (Fig. 6), indicating that participants nonetheless recognized the system's value. We view this tension between usability and intervention depth as an honest design challenge rather than a failure: meaningful psychological support may inevitably involve short-term friction as users engage with unfamiliar cognitive techniques.

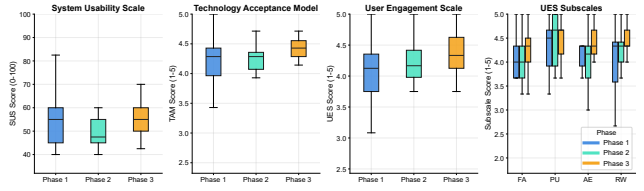


Figure 6: System evaluation metrics by phase.

Qualitative Findings. Exit interviews revealed: (1) proactive buffer turned vague deadline pressure into manageable targets; (2) initial resistance to attributional restructuring (4/16) gave way to strategy internalization; (3) non-academic chat legitimized breaks (11/16); (4) by Phase 3, 8/16 described SoulGrad as a study partner; (5) 10/16 spontaneously transferred reappraisal techniques to situations outside the system.

Table 4: System Evaluation Means (SD) by Phase.

Measure	Phase 1	Phase 2	Phase 3
SUS (0–100)	54.69 (11.23)	49.38 (12.56)	54.69 (10.87)
TAM Usefulness (1–5)	4.12 (0.88)	4.31 (0.87)	4.44 (0.74)
UES-SF Reward (1–5)	4.00 (0.91)	4.18 (0.86)	4.35 (0.70)

5.3 Effects of Preparation Stage

Post-hoc comparison of early (<6 months, $n = 5$) vs. late (>6 months, $n = 11$) preparation revealed that late-stage participants had higher baseline GAD-7 ($M = 11.5$ vs. 7.4) and larger reductions ($M_{diff} = 6.1$ vs. 2.8). This aligns with formative findings that accumulated pressure intensifies both distress and receptivity to support. Subgroups are too small for formal comparison; we report this to motivate future stratified designs.

6 Discussion

We now connect our findings to relevant theoretical frameworks and derive design implications for future ubiquitous systems that integrate LLM-driven mental health support.

6.1 Proactive Intervention as a Paradigm for High-Pressure Contexts

SoulGrad treats affective states as core contextual factors, not secondary outcomes, aligning with recent calls to move beyond passive detection toward actionable sensing in mental health [2]. The three-step attributional restructuring mechanism, grounded in Weiner's Attribution Theory [23], showed encouraging signs of transfer: 10 out of 16 participants spontaneously applied reappraisal strategies in situations outside the system, suggesting that theory-driven restructuring may help cultivate self-regulatory skills over time, though longer deployments are needed to confirm durability. A key design tension concerned intervention timing: users valued proactive support yet demanded absolute control, resolved by the *Do Not Disturb* button and scenario-isolated sensing. We recommend *opt-in* defaults for proactive features: interventions should be offered, not imposed. The front-loaded buffer further relieved relaxation guilt by externally authorizing rest and redistributing tasks within a safety margin.

6.2 Implicit Sensing, Expressive Modalities, and Trust Boundaries

Voice-based venting lowered the barrier to emotional disclosure, consistent with media richness theory, while projective drawing externalized subconscious states that participants could not verbalize. Participants also reported that the novelty of these modalities helped maintain their interest during periods when questionnaire fatigue might otherwise reduce engagement. Participants drew an absolute red line at facial recognition yet accepted smartwatch heart rate monitoring during designated study windows, consistent with privacy calculus theory [11]. This suggests that context-aware mental health systems should adopt scenario-isolated sensing: data collection should be temporally bounded and user-controlled.

6.3 Non-Competitive Social Design

Candidates were caught in a social paradox: profound structural loneliness yet defensive socializing driven by zero-sum competition. The 1+N+2 co-presence model decoupled companionship from comparison by maintaining collective awareness through crowd-level prompts while hiding individual progress. Drawing on parasocial interaction theory [10], system companions provided unconditional emotional receptacles without the reciprocity burden of human relationships. Prescribed breaks alleviated rest guilt by reframing pauses as algorithmically authorized, a strategy we term *externalized permission*. These findings suggest that context-adaptive social design can help balance the need for connection with psychological safety in competitive environments.

6.4 Limitations

This study has several limitations. It is a single-group pilot ($N = 16$, 16 days), so causal attribution is not possible. Multiple comparisons were not corrected, and intervention thresholds were heuristically set rather than individually calibrated. Future work should conduct an RCT with a larger, more diverse sample, deploy the system over a full preparation lifecycle, and benchmark LLM safety guardrails against standardized evaluation datasets.

7 Conclusion

This paper presented SoulGrad, an LLM-driven system for supporting mental wellbeing during high-stakes exam preparation. Through a mixed-method formative study and a 16-day pilot evaluation, we explored the feasibility of coupling behavioral sensing with proactive psychological intervention, while critically examining the ethical tensions of deploying LLM-driven systems for vulnerable populations. As the UbiComp community increasingly integrates large language models into mental health systems, we must collectively develop standards for safety evaluation, transparent reporting, and responsible deployment. We offer this work as a starting point for workshop discussion on navigating the opportunities and risks of LLM-driven mental health support.

Acknowledgments

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